Unit - 5

Prompt Engineering – What is a Prompt?

A prompt is a set of instructions, questions, or information given to an AI model to generate a desired response or perform a specific task.

In simple words, a prompt is the input or instruction that a user gives to an AI model.

1). Examples of Prompts

- **Question:** What is Artificial Intelligence?
- **Instruction:** Write a C program to calculate the factorial of a number.
- **Content Generation:** Write a paragraph on environmental pollution.
- **Image Generation:** Generate an image of a futuristic city at night.
- **Translation:** Translate this paragraph from Hindi to English.

2). Components of a Prompt

- **Instruction:** Specifies what the AI model should do.
- **Context:** Provides background information about the task.
- **Input Data:** Provides the text, question, or information to be processed.
- **Output Format:** Specifies how the response should be presented.

3). Example of a Good Prompt

Prompt: "Explain Artificial Intelligence in simple English for B.Tech first-year students in 100 words with two examples."

Explanation:

- Task: Explain Artificial Intelligence.

- Language: Simple English.
- Audience: B.Tech first-year students.
- Length: 100 words.
- Examples: Two examples.

4). Characteristics of a Good Prompt

- Clear and specific instructions.
- Provides sufficient context.
- Uses simple and understandable language.
- Clearly defines the expected output.
- Avoids unnecessary or confusing information.

5). Importance of Prompts

- Improves the accuracy of AI responses.
- Helps generate relevant information.
- Reduces misunderstanding.
- Saves time and effort.
- Controls the format and style of output.

Why Prompt Quality Matters

Prompt quality refers to how clearly, specifically, and effectively a user provides instructions to an AI model. A good-quality prompt helps the AI generate accurate, relevant, and useful responses.

The same AI model can produce very different results depending on the quality of the prompt.

1). Importance of Prompt Quality

- **Improves Accuracy:** Clear instructions help the AI generate correct and relevant answers.
- **Provides Better Results:** Specific prompts produce more useful and detailed responses.
- **Reduces Confusion:** Proper instructions minimise misunderstanding.
- **Saves Time:** Good prompts reduce the need for repeated instructions.
- **Controls Output Format:** Prompts can specify the required language, length, structure, and style.

2). Example: Same AI, Different Prompts

Prompt 1 (Poor Quality):

"Explain Python."

Result: The AI may provide a general explanation of Python without considering the user's level or requirements.

Prompt 2 (Good Quality):

"Explain Python programming in simple English for B.Tech first-year students. Include its definition, features, applications, and one example."

Result: The AI provides a structured, detailed, and exam-oriented explanation.

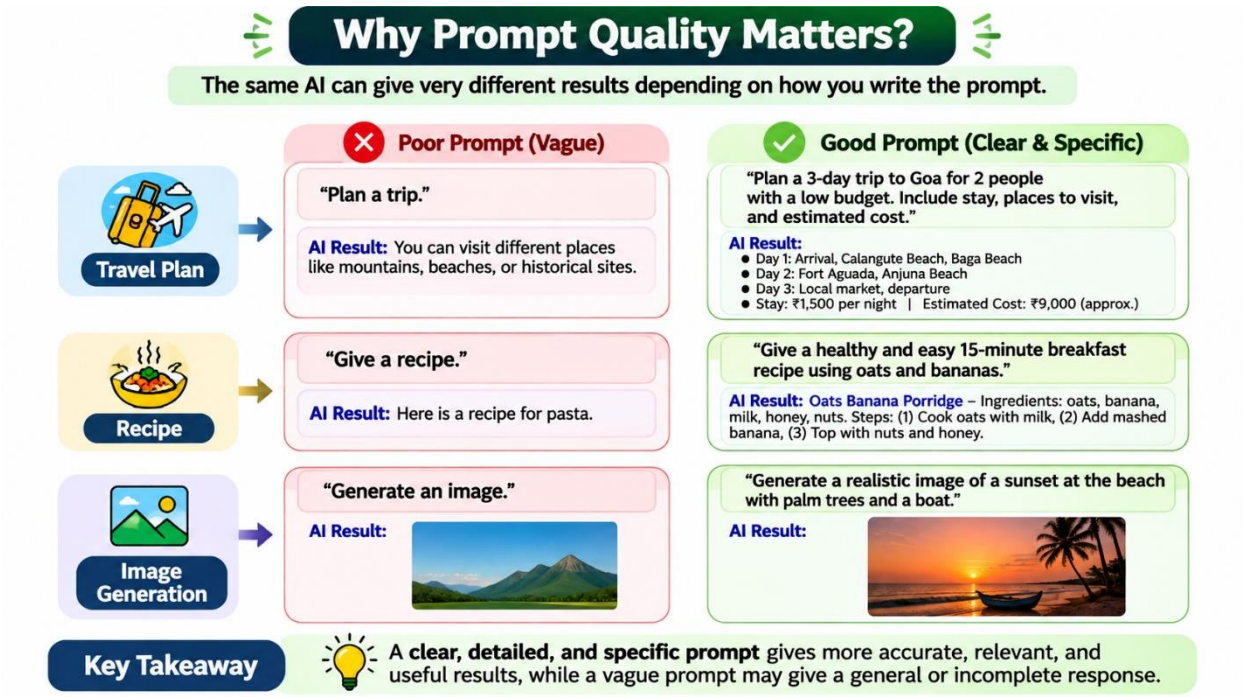
3). Factors Affecting Prompt Quality

- Clarity of instructions.
- Specificity of the task.
- Sufficient background information.
- Clear output requirements.

- Appropriate examples and context.

4). Characteristics of a Good Prompt

- Clear and simple language.
- Specific and meaningful instructions.
- Relevant background information.
- Clearly defined expected output.
- No unnecessary information.



Principles of Good Prompting

Good prompting is the process of writing clear, specific, and well-structured instructions to an AI model to obtain accurate, relevant, and useful responses.

A prompt acts as a communication link between the user and the AI model. The quality of the prompt directly affects the quality of the generated response.

The five main principles of good prompting are:

1. Clarity

Clarity means providing instructions in simple, clear, and understandable language so that the AI model can easily understand what the user wants.

A clear prompt avoids confusion and helps the AI generate the required response without misunderstanding the task.

Important Points:

- Use simple and understandable language.
- Give direct and meaningful instructions.
- Avoid confusing or ambiguous words.
- Clearly mention the expected task.
- Avoid combining unrelated instructions.

Example:

Poor Prompt

"Tell me something about fitness."

Good Prompt

"Explain five health benefits of daily exercise in simple English."

Result: The good prompt clearly specifies what information is required and how it should be explained.

2. Context

Context means providing background information, relevant details, or the situation related to a task so that the AI model can understand the user's requirements properly.

Context helps AI generate responses according to the user's needs instead of providing general information.

Important Points:

- Provide necessary background information.
- Explain the purpose of the task.
- Mention the target audience.
- Include relevant details about the situation.
- Provide previous information when required.

Example:

Poor Prompt

"Suggest a diet plan."

Good Prompt

"I am a 25-year-old vegetarian male who wants to gain weight. Suggest an affordable daily diet plan using commonly available Indian foods."

Result: The good prompt provides personal requirements and background information, allowing AI to generate a more relevant diet plan.

3. Specificity

Specificity means providing detailed and exact instructions about what the AI model should generate. It defines the scope, requirements, limitations, and expected details of the task.

A specific prompt reduces unnecessary information and helps the AI focus on the actual requirement.

Important Points:

- Clearly define the task.
- Mention the required topics.

- Specify the length or word limit.
- Mention important conditions or restrictions.
- Avoid vague and general instructions.

Example:

Poor Prompt

"Plan a trip to Goa."

Good Prompt

"Plan a 3-day trip to Goa for two people within a budget of ₹15,000. Include accommodation, food, transportation, sightseeing places, and estimated expenses."

Result: The good prompt provides exact requirements, making the travel plan more practical and useful.

4. Role Assignment

Role assignment means instructing an AI model to act as a particular professional, expert, teacher, or specialist while answering a question.

It helps the AI generate responses from a specific perspective and maintain an appropriate level of knowledge, language, and communication style.

Important Points:

- Assign a suitable role to the AI.
- Choose a role according to the task.
- Mention the expected expertise.
- Specify the target audience.
- Define the communication style if required.

Example:

Poor Prompt

"Explain healthy eating."

Good Prompt

"Act as a professional nutritionist and explain the importance of a balanced diet, including proteins, carbohydrates, fats, vitamins, and minerals in simple language."

Result: The good prompt assigns an expert role and clearly specifies the information required.

5. Format Instruction

Format instruction means specifying the structure, layout, and presentation style in which the AI model should generate its response.

It helps users receive information in an organised format that is easy to read, understand, and use.

Important Points:

- Specify whether the answer should be in paragraphs, bullet points, or tables.
- Mention the required language.
- Specify the word limit or answer length.
- Mention whether examples are required.
- Define the output structure.

Example:

Poor Prompt

"Compare electric cars and petrol cars."

Good Prompt

"Compare electric cars and petrol cars in a table using five points: cost, maintenance, fuel, environmental impact, and performance. Use simple English."

Result: The good prompt specifies the required format and comparison points, producing an organised response.

6. Combined Example of Good Prompting

A good prompt can combine all five principles to obtain a better AI response.

Example Prompt:

"Act as a professional travel planner. I want to visit Jaipur with my family for two days. Suggest an affordable travel plan within ₹10,000. Include famous tourist places, food, accommodation, and transportation. Present the answer in a table using simple English."

Principle	**How it is used**
Clarity	Clearly asks for a travel plan.
Context	Mentions a family trip to Jaipur.
Specificity	Specifies two days and ₹10,000 budget.
Role Assignment	Assigns the role of a professional travel planner.
Format Instruction	Requests a table in simple English.